THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 4: **Quadratic Equations** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 29	$(\alpha+\beta)^2-2\alpha\beta$
2	**(a)** 24	$(\alpha-\beta)^2=(\alpha+\beta)^2-4\alpha\beta$
3	**(c)** 6	$D=0$ gives $k^2=4m$
4	**(b)** 2	Substitute the root
5	**(c)** $k\le 3$ and $k\neq 0$	$D=36-12k\ge 0$
6	**(b)** $\frac{5}{6}$	$\frac{\alpha+\beta}{\alpha\beta}$
7	**(d)** 53	$(\alpha+\beta)^2-2\alpha\beta$
8	**(d)** 32	$(\alpha-\beta)^2=(\alpha+\beta)^2-4\alpha\beta$
9	**(a)** Both A and R are true and R is the correct explanation of A.	$D=9-40=-31<0$, so there are no real roots, as stated in R.
10	**(c)** A is true but R is false.	$D=36-36=0$ gives equal roots (3 and 3): A true; R is false.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$D=(4)^2-4k(1)=0$ (1); $k=4$ (1).	1+1
12	$\alpha+\beta=6$, $\alpha\beta=5$ (1); $\alpha^2+\beta^2=6^2-2(5)=26$ (1).	1+1
13	Let number x: $x+\frac{1}{x}=\frac{5}{2}$ (1); $x=2$ or $x=\frac{1}{2}$ (1).	1+1
14	$D=(6)^2-4k(3)=0$ (1); $k=3$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	LHS $=\frac{2(x^2+1)}{x^2-1}$ (1); cross-multiplying: $3(x^2+1)=5(x^2-1)$ (1); $x^2=4$ so $x=\pm 2$ (1).	1+1+1
16	$D=25-4k\ge 0$ (1); $k\le\frac{25}{4}$ (1); greatest integer $k=6$ (1).	1+1+1
17	Integers x, $x+2$: $x^2+(x+2)^2=74$ (1); $x^2+2x-35=0$ (1); $x=5$; integers 5 and 7 (1).	1+1+1
	OR: LHS $=\frac{2(x^2+4)}{x^2-4}$ (1); cross-multiplying: $5(x^2+4)=13(x^2-4)$ (1); $x^2=9$ so $x=\pm 3$ (1).	1+1+1
18	$D=49-4k\ge 0$ (1); $k\le\frac{49}{4}$ (1); greatest integer $k=12$ (1).	1+1+1
	OR: Integers x, $x+2$: $x^2+(x+2)^2=130$ (1); $x^2+2x-63=0$ (1); $x=7$; integers 7 and 9 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $x(x+3)=180\Rightarrow x^2+3x-180=0$ (1). (ii) $(x+15)(x-12)=0$, so $x=12$ (rejecting -15) (1). (iii) Breadth 12 m, length 15 m (1). (iv) Perimeter $=2(15+12)=54$ m; wire $=3\times 54=162$ m (1); cost $=162\times 20=₹3240$ (1).	1+1+1+2
20	Smaller tap x h, larger $x-4$ h: $\frac{1}{x}+\frac{1}{x-4}=\frac{1}{\frac{24}{5}}$ (2); simplify to a quadratic and solve, rejecting the root less than 4: $x=12$ (2); times 12 h and 8 h (1).	2+2+1
	OR: Boat speed x: $\frac{60}{x-4}+\frac{60}{x+4}=8$ (1); $60(2x)=8(x^2-16)$ (1); roots of the resulting quadratic: $x=16$ or negative (1); speed 16 km/h (1).	1+1+1+1

Mistakes to avoid:

- Writing the formula as $\frac{-b\pm\sqrt{b^2+4ac}}{2a}$ or dividing only part of the numerator by $2a$.
- Forgetting the sign of b: for $x^2-5x+6=0$, $-b=+5$.
- Not shifting everything to one side to get the standard form $ax^2+bx+c=0$ before using D or factorising.
- Cancelling a common factor containing x from both sides, which loses a root (e.g. $x(x-2)=3x$).

- Not rejecting a negative root for length, speed, age or time in word problems.
- Saying $D = 0$ means no roots; it means two *equal* real roots.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 2

Chapter 4: **Quadratic Equations** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** $k \leq 2$ and $k \neq 0$	$D = 16 - 8k \geq 0$
2	**(c)** $\frac{7}{12}$	$\frac{\alpha+\beta}{\alpha\beta}$
3	**(b)** 58	$(\alpha+\beta)^2 - 2\alpha\beta$
4	**(b)** 40	$(\alpha-\beta)^2 = (\alpha+\beta)^2 - 4\alpha\beta$
5	**(b)** 10	$D = 0$ gives $k^2 = 4m$
6	**(a)** -2	Substitute the root
7	**(b)** $k \leq 4$ and $k \neq 0$	$D = 64 - 16k \geq 0$
8	**(c)** $\frac{9}{20}$	$\frac{\alpha+\beta}{\alpha\beta}$
9	**(d)** A is false but R is true.	$D = 16 - 20 = -4 < 0$, so there are no real roots: A false; R is a true statement.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	$(x-2)(x-3) = 0$ gives 2 and 3. Both statements are true, but R does not explain why these particular numbers are roots.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$\alpha+\beta = 8$, $\alpha\beta = 7$ (1); $\alpha^2+\beta^2 = 8^2 - 2(7) = 50$ (1).	1+1
12	Let number x: $x + \frac{1}{x} = \frac{10}{3}$ (1); $x = 3$ or $x = \frac{1}{3}$ (1).	1+1
13	$D = (8)^2 - 4k(8) = 0$ (1); $k = 2$ (1).	1+1
14	$\alpha+\beta = 9$, $\alpha\beta = 8$ (1); $\alpha^2+\beta^2 = 9^2 - 2(8) = 65$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	LHS $= \frac{2(x^2+1)}{x^2-1}$ (1); cross-multiplying: $8(x^2+1) = 10(x^2-1)$ (1); $x^2 = 9$ so $x = \pm 3$ (1).	1+1+1
16	$D = 81 - 4k \geq 0$ (1); $k \leq \frac{81}{4}$ (1); greatest integer $k = 20$ (1).	1+1+1
17	Integers x, $x+2$: $x^2 + (x+2)^2 = 202$ (1); $x^2 + 2x - 99 = 0$ (1); $x = 9$; integers 9 and 11 (1).	1+1+1
	OR: LHS $= \frac{2(x^2+9)}{x^2-9}$ (1); cross-multiplying: $16(x^2+9) = 34(x^2-9)$ (1); $x^2 = 25$ so $x = \pm 5$ (1).	1+1+1
18	$D = 121 - 4k \geq 0$ (1); $k \leq \frac{121}{4}$ (1); greatest integer $k = 30$ (1).	1+1+1
	OR: Integers x, $x+2$: $x^2 + (x+2)^2 = 290$ (1); $x^2 + 2x - 143 = 0$ (1); $x = 11$; integers 11 and 13 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $\frac{360}{x} - \frac{360}{x+5} = 1$ (1). (ii) $360(x+5) - 360x = x(x+5) \Rightarrow x^2 + 5x - 1800 = 0$; $D = 25 + 7200 = 7225 = 85^2$ (1). (iii) $x = \frac{-5 \pm 85}{2}$, so $x = 40$ km/h (speed cannot be negative) (1). (iv) Time at 40 km/h $= 9$ h (1); at 45 km/h $= 8$ h; difference $= 1$ h, verified (1).	1+1+1+2
20	Smaller tap x h, larger $x-3$ h: $\frac{1}{x} + \frac{1}{x-3} = \frac{1}{\frac{18}{5}}$ (2); simplify to a quadratic and solve, rejecting the root less than 3: $x = 9$ (2); times 9 h and 6 h (1).	2+2+1
	OR: Speed x: $\frac{1080}{x} - \frac{1080}{x+5} = 3$ (1); $5400 = 3x(x+5)$ so $x^2 + 5x - 1800 = 0$ (2); $(x-40)(x+45) = 0$ (1); speed 40 km/h (1).	1+2+1+1

Mistakes to avoid:

- Writing the formula as $\frac{-b \pm \sqrt{b^2+4ac}}{2a}$ or dividing only part of the numerator by $2a$.

- Forgetting the sign of b: for $x^2 - 5x + 6 = 0$, $-b = +5$.
- Not shifting everything to one side to get the standard form $ax^2 + bx + c = 0$ before using D or factorising.
- Cancelling a common factor containing x from both sides, which loses a root (e.g. $x(x - 2) = 3x$).
- Not rejecting a negative root for length, speed, age or time in word problems.
- Saying $D = 0$ means no roots; it means two *equal* real roots.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 3

Chapter 4: **Quadratic Equations** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 12	$D = 0$ gives $k^2 = 4m$
2	**(b)** 2	Substitute the root
3	**(c)** $k \leq 5$ and $k \neq 0$	$D = 100 - 20k \geq 0$
4	**(b)** $\frac{8}{15}$	$\frac{\alpha+\beta}{\alpha\beta}$
5	**(c)** 40	$(\alpha+\beta)^2 - 2\alpha\beta$
6	**(c)** 65	$(\alpha-\beta)^2 = (\alpha+\beta)^2 - 4\alpha\beta$
7	**(a)** 14	$D = 0$ gives $k^2 = 4m$
8	**(a)** 2	Substitute the root
9	**(a)** Both A and R are true and R is the correct explanation of A.	Expanding: $x^3 + 6x^2 + 12x + 8 = x^3 - 4 \Rightarrow 6x^2 + 12x + 12 = 0$ which is of the form in R.
10	**(c)** A is true but R is false.	$2x^2 - 7x + 3 = (2x-1)(x-3)$ so A is true; the formula has $b^2 - 4ac$, so R is false.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Let number x: $x + \frac{1}{x} = \frac{17}{4}$ (1); $x = 4$ or $x = \frac{1}{4}$ (1).	1+1
12	$D = (10)^2 - 4k(5) = 0$ (1); $k = 5$ (1).	1+1
13	$\alpha + \beta = 10$, $\alpha\beta = 9$ (1); $\alpha^2 + \beta^2 = 10^2 - 2(9) = 82$ (1).	1+1
14	Let number x: $x + \frac{1}{x} = \frac{26}{5}$ (1); $x = 5$ or $x = \frac{1}{5}$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	LHS $= \frac{2(x^2+4)}{x^2-4}$ (1); cross-multiplying: $21(x^2+4) = 29(x^2-4)$ (1); $x^2 = 25$ so $x = \pm 5$ (1).	1+1+1
16	$D = 169 - 4k \geq 0$ (1); $k \leq \frac{169}{4}$ (1); greatest integer $k = 42$ (1).	1+1+1
17	Integers x, $x+2$: $x^2 + (x+2)^2 = 394$ (1); $x^2 + 2x - 195 = 0$ (1); $x = 13$; integers 13 and 15 (1).	1+1+1
	OR: LHS $= \frac{2(x^2+1)}{x^2-1}$ (1); cross-multiplying: $15(x^2+1) = 17(x^2-1)$ (1); $x^2 = 16$ so $x = \pm 4$ (1).	1+1+1
18	$D = 225 - 4k \geq 0$ (1); $k \leq \frac{225}{4}$ (1); greatest integer $k = 56$ (1).	1+1+1
	OR: Integers x, $x+2$: $x^2 + (x+2)^2 = 514$ (1); $x^2 + 2x - 255 = 0$ (1); $x = 15$; integers 15 and 17 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $D = 36 - 24 = 12$ (1). (ii) $D = 48 - 48 = 0$, so roots are real and equal (1). (iii) $D = 4 - 20 = -16 < 0$, so no real roots (1). (iv) Equal roots need $D = k^2 - 4(2)(3) = 0$ (1), so $k^2 = 24$, $k = \pm 2\sqrt{6}$ (1).	1+1+1+2
20	Boat speed x: $\frac{40}{x-5} + \frac{40}{x+5} = 6$ (1); $40(2x) = 6(x^2-25)$ (1); roots of the resulting quadratic: $x = 15$ or negative (1); speed 15 km/h (1).	1+1+1+1
	OR: Speed x: $\frac{720}{x} - \frac{720}{x+4} = 2$ (1); $2880 = 2x(x+4)$ so $x^2 + 4x - 1440 = 0$ (2); $(x-36)(x+40) = 0$ (1); speed 36 km/h (1).	1+2+1+1

Mistakes to avoid:

- Writing the formula as $\frac{-b \pm \sqrt{b^2+4ac}}{2a}$ or dividing only part of the numerator by $2a$.
- Forgetting the sign of b: for $x^2 - 5x + 6 = 0$, $-b = +5$.
- Not shifting everything to one side to get the standard form $ax^2 + bx + c = 0$ before using D or factorising.
- Cancelling a common factor containing x from both sides, which loses a root (e.g. $x(x-2) = 3x$).

- Not rejecting a negative root for length, speed, age or time in word problems.
- Saying $D = 0$ means no roots; it means two *equal* real roots.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND